Btech Sem 5
Compiler Design Assignment 1

1. What is compiler? What are the factors that effect design of compiler and functions of compiler.

2. Explain Phases of compiler in detail with example (a=a+b*c*2)

3. Convert the following NFA into an equivalent DFA and identify all reachable states.

- States: {q0, q1, q2, q3}
- Alphabet: {0,1}
- Start State: q0
- Final State: {q2, q3}

State	0	1
q0	{q0, q1}	{q0}
q1	{q2}	{q3}
q2	{q2}	{q2, q3}
q3	∅	{q3}

4. Convert NFA-^ to NFA and then DFA.
States: {q0, q1, q2}

- Alphabet: {a, b}
- Start State: q0
- Final State: {q2}
- Transitions:

- q0 --ε→ q1
- q1 --a→ q1
- q1 --b→ q2
- q2 --a→ q2

5. Define parser? What are the techniques of parser.

6. Check following grammer is LL(1) or not? (^ - null)
S -> aBDh
B -> cC
C -> bC / ^ / e
D -> E
E -> g / ^

F -> f /^

7. Remove left recursion:
A-> Abd / Aa / a
B->Be/b

8. Following grammar LL(1) or not?
S->1AB / ^
A->1AC/0C
B->0S
C->1
Also parse the string ‘110110’.

9.Following grammar LL(1) or not?
S ⟶ AB
A ⟶ +|-|ε
B ⟶ digit | B digit
C ⟶ B